

FINTECH 540 - Machine Learning for Fintech

Fall Semester 2025

Second Project Checkpoint



- ▶ We've noticed that volatility tends to spike in certain periods, which might reflect shifts in market conditions like bull or bear phases. We're thinking about using models that can detect these changes, but we're not sure if that's too complex for this project or if a simpler approach would be better. What do you think?
- ▶ How can we identify regime boundaries? If we use a method like fixed threshold, it may not adapt to market evolution. Should we use statistical methods like HMM or Gaussian mixture to let data determine regimes automatically?
 - ▶ *Expanding-window quantiles on (log) RV (e.g., 0-40% = Low, 40-80% = Med, 80-100% = High); add 3-day hysteresis.*

- ▶ We are considering whether or not to include VIX data for predicting abnormal regimes. Given that it measures expected market-wide volatility based on S&P500 index options rather than on single stocks, we're wondering if using this data would be helpful or not since our given volatility data is per stock.
 - ▶ *VIX can be a factor for common market volatility.*
- ▶ How should we split train/test datasets for the time series data? We need to ensure all the regimes are represented in both datasets while maintaining temporal order.
 - ▶ *Rolling windows are an option.*

- ▶ Feature Selection and Interpretation: We are presently investigating which features are most effective in predicting volatility. The dataset contains a variety of financial variables, but we don't yet know which ones contribute the most to model performance and interpretability.
 - ▶ *SHAP*
- ▶ We are considering whether to incorporate broader economic indicators such as GDP growth, inflation rates, or interest rate changes into our volatility model. Our question is whether these macro-level variables typically enhance predictive power in volatility forecasting, or if they introduce unnecessary noise given our current dataset?
 - ▶ *You may encounter a mixed frequency data problem.*

- ▶ Data Cleaning and Outlier Handling: During our EDA, we discovered variables with missing data or severe spikes. We are reviewing several methods for cleaning or normalizing these data sets without losing essential trends.
- ▶ What should we do to deal with the financial data with different start dates?
- ▶ Should we verify whether missing data points or non-trading days were interpolated, dropped, or forward-filled?

- ▶ If doing MEV detection, should we use raw values or normalized/scaled inputs for MEV detection? Does token value inflation matter?
- ▶ Some addresses are labeled MEV; others aren't. How should semi-supervision inform the target setup?

- ▶ How should we best handle the class imbalance of fraudulent and regular transactions? We know there will be far more regular transactions than fraudulent ones.
- ▶ In which range of values (both cost and frequency) for transfers might be a sign of a fraudulent transaction? As in, what cutoffs or what's an example of these fraudulent transactions, so we have a baseline of what we're looking at.

- ▶ Model Framework Decision: We're discussing whether to use regression-based forecasting models (e.g., Linear Regression, Random Forest) or time-series methods (e.g., ARIMA, LSTM). We'd like your thoughts on which path best matches the course requirements.

- ▶ How to account for the impact of extreme black swan events, such as how to predict something like another financial crisis occurring?

Further Questions? Comments?